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System for generating product recommendations using biometric data

Abstract

Systems and methods for facilitating product preferences and/or product recommendations are disclosed. Biometric data of the subject are taken into account by the systems and methods in order to determine product preferences and/or product recommendations. Other factors of the subject, although optional, may be also taken into account while determining product preferences and/or product recommendations for a subject. These product preferences or recommendations can be presented to the subject, either automatically via a display device or with assistance from a product consultant. To obtain the biometric data, the subject will be exposed to a number of stimuli, such as fragrance/scent stimuli. Biometric data will then be collected from the subject based on her response to this fragrance/scent stimuli. In some examples, the collected biometric data relates to a subject's Electrodermal Resistance (EDR) or Electrodermal Activity (EDA).

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/046,410, filed Jun. 30, 2020, the disclosures of which are incorporated herein in their entirety.

FIELD OF THE DISCLOSURE

(1) Embodiments of the present disclosures relate generally to the use of biometric data as an aid in product recommendations and/or selection. In some embodiments, the biometric data is utilized to determine product preferences of a user. In some of these embodiments, the product preferences are used to aid in product recommendations and/or selection.

SUMMARY OF THE DISCLOSURE

(2) Systems and methods for facilitating product preferences and/or product recommendations are disclosed herein. In embodiments described below, biometric data of the subject are taken into account by the systems and methods in order to determine product preferences and/or product recommendations. In some embodiments, other factors of the subject, although optional, may be also taken into account while determining product preferences and/or product recommendations for a subject. These product preferences or recommendations can be presented to the subject, either automatically via a display device or with assistance from a product consultant.

(3) To obtain the biometric data in some embodiments, the subject will be exposed to a number of stimuli, such as fragrance/scent stimuli. Biometric data will then be collected from the subject based on her response to this fragrance/scent stimuli. In some examples, the collected biometric data relates to a subject's Electrodermal Resistance (EDR) or Electrodermal Activity (EDA).

(4) For example, in some embodiments, the systems and methods detect an electrodermal response based on the response to an olfactory stimulus. In this regard, the system and methods detect a real-time cognitive process associated with an olfactory stimulus; detect a variation in an electrical characteristics of the skin associated with a response to one or more fragrance accords; detect a skin conductance based on a response to an olfactory stimulus; detect a change in a skin potential

associated with a response to an olfactory stimulus; or detect conductivity fluctuations indicative of a response to an olfactory stimulus.

(5) In accordance with an aspect of the present disclosure, a system is provided that comprises in an embodiment a plurality of sensors configured to sense an electrodermal response of a subject based on a response to an olfactory stimulus, and one or more engines configured to: receive the electrodermal response of a subject as Galvanic Skin Response (GSR) signals; process the GSR signals to generate GSR data; and generate a product recommendation based at least on said GSR data.

(6) In some embodiments, the one or more engines are housed in a mobile computing device.

(7) In some embodiments, the generated GSR data is represented as an image.

(8) In some embodiments, the one or more engines are further configured to generate preferred characteristic parameters of the olfactory stimulus based on the GSR data and to generate the product recommendation based on the preferred characteristic parameters.

(9) In some embodiments, the one or more engines are configured to determine a product recommendation by comparing data indicative of the generated preferred characteristic parameters to product data accessible by the one or more engines.

(10) In some embodiments, the olfactory stimulus is a fragrance and wherein the data indicative of the generated preferred characteristic parameters includes a fragrance profile.

(11) In some embodiments, the fragrance profile is presented to the subject as the product recommendation.

(12) In some embodiments, the generated preferred characteristic parameters represent notes of the fragrance. In some of these embodiments, the product recommendation is generated by comparing the fragrance profile with a set of fragrance profiles representing, respectively, a set of fragrances, the set of fragrance profiles accessible by the one or more engines.

(13) In some embodiments, the system further comprising a camera configured to capture data indicative of the emotional state of the subject based on the response to the olfactory stimulus, wherein the product recommendation is generated based at least on said GSR data and said data indicative of the emotional state of the subject.

(14) In some embodiments, the one or more engines include processing circuitry configured to: detect a real-time cognitive process associated with an olfactory stimulus, detect a variation in an electrical characteristics of the skin associated with a response to one or more fragrance accords, detect a skin conductance based on a response to an olfactory stimulus, detect a change in a skin potential associated with a response to an olfactory stimulus, or detect conductivity fluctuations indicative of a response to an olfactory stimulus.

(15) In some embodiments, the plurality of sensors and/or the one or more engines form a scent response unit includes processing circuitry configured to detect one of: a real-time cognitive process associated with an olfactory stimulus; a variation in an electrical characteristics of the skin associated with a response to one or more fragrance accords; detect a skin conductance based on a response to an olfactory stimulus; a change in a skin potential associated with a response to an olfactory stimulus; or conductivity fluctuations indicative of a response to an olfactory stimulus.

(16) In some embodiments, the olfactory stimulus includes a scent. In some of these embodiments, the one or more engines form a perfume selection unit that includes one of the following: processing circuitry configured to generate one or more virtual instances of a fragrance subset based on at least one input associated with an electrodermal activity measurand; processing circuitry configured to generate one or more virtual instances of a fragrance subset based on at least one input associated with a sympathetic skin response; processing circuitry configured to generate one or more instances of a degree of desirability score or a likeability measure; processing circuitry configured to generate one or more instances of a scent strength; processing circuitry configured to generate one or more instances of aromatic compound concertation; or processing circuitry configured to generate one or more instances of base notes, top notes or middle notes in a fragrance

composition.

(17) In accordance with another aspect of the present disclosure, a system is provided that comprises a scent response unit and a perfume selection unit. In some embodiments, the scent response unit includes processing circuitry configured to detect an electrodermal response based on a response to an olfactory stimulus. In some embodiments, the perfume selection unit including processing circuitry configured to generate one or more virtual instances of a fragrance subset based on at least one input associated with the electrodermal response.

(18) In some embodiments, the scent response unit includes processing circuitry configured to: detect a real-time cognitive process associated with an olfactory stimulus; detect a variation in an electrical characteristics of the skin associated with a response to one or more fragrance accords; detect a skin conductance based on a response to an olfactory stimulus; detect a change in a skin potential associated with a response to an olfactory stimulus; or detect conductivity fluctuations indicative of a response to an olfactory stimulus.

(19) In some embodiments, the perfume selection unit further includes one of: processing circuitry configured to generate one or more virtual instances of a fragrance subset based on at least one input associated with an electrodermal activity measurand; processing circuitry configured to generate one or more virtual instances of a fragrance subset based on at least one input associated with a sympathetic skin response; processing circuitry configured to generate one or more instances of a degree of desirability score or a likeability measure; processing circuitry configured to generate one or more instances of a scent strength; processing circuitry configured to generate one or more instances of aromatic compound concertation; or processing circuitry configured to generate one or more instances of base notes, top notes or middle notes in a fragrance composition.

(20) In accordance with another aspect of the present disclosure, a method is provided for recommending a product to a subject. In an embodiment, the method includes obtaining Galvanic Skin Response (GSR) data of a subject based on exposure to an olfactory stimulus, analyzing the GSR data; and recommending a product to the subject based on at least the analyzed GSR data.

(21) In some embodiments, recommending a product to the subject based on the analyzed GSR data includes either presenting a fragrance name to the subject or presenting a fragrance profile to the subject.

(22) In some embodiments, the method further comprises obtaining questionnaire data of the subject indicative of a preference of a characteristic parameter of a product. In some embodiments, the recommendation of the product to the subject is based on the analyzed biometric data and the questionnaire data.

(23) In some embodiments, the olfactory stimulus is a scent, and wherein the product is a perfume.

(24) In some embodiments, recommending a product includes generating a fragrance profile based on the GSR data and presenting the fragrance profile to the subject.

(25) In some embodiments, recommending a product includes generating a fragrance profile based on the GSR data, comparing the fragrance profile with a set of fragrance profiles representing, respectively, a set of fragrances to select a fragrance from the set of fragrances having a fragrance profile most similar to the generated fragrance profile, and presenting the selected fragrance to the subject.

(26) In some embodiments, the method further comprises obtaining data indicative of the emotional state of the subject based on the exposure to the olfactory stimulus. In some embodiments, the recommendation of the product to the subject is based on the analyzed GSR data and said data indicative of the emotional state of the subject.

(27) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

Description

DESCRIPTION OF THE DRAWINGS

(1) The foregoing aspects and many of the attendant advantages of the disclosed subject matter will become more readily appreciated as the same become better understood by reference to the following detailed description, when taken in conjunction with the accompanying drawings, wherein:

(2) FIG. 1 is a schematic diagram that illustrates a non-limiting example of a system for generating and providing product, such as fragrance, recommendations to a subject according to an aspect of the present disclosure;

(3) FIG. 2 depicts one example of a GSR unit suitable for use in the system of FIG. 1;

(4) FIG. 3 is a block diagram that illustrates a non-limiting example of a mobile computing device suitable for use in the system of FIG. 1;

(5) FIG. 4 is a block diagram that illustrates a non-limiting example of a server computing device suitable for use in the system of FIG. 1;

(6) FIG. 5 is a block diagram that illustrates a non-limiting example of a computing device appropriate for use as a computing device with embodiments of the present disclosure;

(7) FIG. 6 is a flowchart that illustrates a non-limiting example of a method of generating and providing product recommendations to a subject according to an aspect of the present disclosure;

(8) FIG. 7 is a flowchart that illustrates another non-limiting example of a method of generating and providing product recommendations to a subject according to an aspect of the present disclosure;

(9) FIG. 8 is an example of a fragrance wheel;

(10) FIG. 9 is one example of a skin conductance graph or GSR graph generated as a result of processing collected GSR signals in accordance with the present disclosure;

(11) FIG. 10 is one example of a note diagram generated by the system and presented to the subject in accordance with an aspect of the present disclosure; and

(12) FIG. 11 illustrates an example of a question, depicting one or more scenes, that is generated by the user interface engine and presented to the subject.

DETAILED DESCRIPTION

(13) In order to provide recommendations for cosmetics, most existing techniques merely attempt to directly discern a subject's product preferences. Some techniques may attempt to determine a subject's product preferences based on the subject's stated preferences for a characteristic of the product, such as smell, color, finish, feel, etc. However, this technique produces sub-optimal recommendations, at least because only explicitly stated subject preferences are taken into account. Even in the presence of explicitly stated subject preferences, other factors (e.g., physical reactions (subconscious or conscious), personality traits, etc.), may also influence what products a given subject will prefer.

(14) In that regard, in some embodiments of the present disclosure, biometric data of the subject are taken into account while determining product preferences and/or product recommendations. For example, physiological response data can be collected and used for determining product recommendations. In other embodiments, other factors of the subject, although optional, may be also taken into account while determining product preferences and/or product recommendations for a subject. These product preferences or recommendations are then represented to the subject, either automatically via a display device or with assistance from a product consultant.

(15) The examples described throughout the disclosure relate to recommendations for a fragrance, such a perfume or cologne. It will be appreciated that the techniques and methodologies of the present disclosure transcend product types, and thus, can be used to provide recommendations to the subject for products other than fragrances.

(16) In the examples described below, a subject will be exposed to a number of fragrance/scent stimuli. Biometric data will then be collected from the subject based on her physiological response to this fragrance/scent stimuli. In some embodiments, the collected biometric data relates to a subject's Electrodermal Resistance (EDR) or Electrodermal Activity (EDA). For example, a change in galvanic skin response (GSR) activity in response to a stimulus is referred to as an Event-Related Skin Conductance Response (ER-SCR), and can provide information about emotional arousal to the stimulus. Also, as may be used in the following, an event-related potential is the measured brain response resulting from a specific sensory, cognitive, or motor event.

(17) With this biometric data, a computer system will, for example, recommend either a specific fragrance or will develop a fragrance profile from which a recommendation can be made with the assistance of, for example, a technician or fragrance consultant. In other embodiments, the computer system will use the biometric data in conjunction with optional data, such as data obtained from a questionnaire, image data of the subject referred later in the description as facial data, historical purchase data of the subject, etc., in order to present a product recommendation to the subject.

(18) In some embodiments described herein, fragrances/scents presented to the subject may include two or more notes of an accord. Generally, an accord is a scent made up of several perfume notes, ingredients, etc., that blend together to form a distinct fragrance. For example, an accord usually includes a number of notes. Notes are descriptors of scents that can be sensed, and include base notes, middle or heart notes, and top or head notes.

(19) These scent descriptors or notes are well known and widely use to describe the odor character (e.g., a characteristic parameter) of a fragrance.

(20) Notes are typically chosen from fragrance families. Generally, fragrance families can be visually presented by a fragrance wheel. One example of a fragrance wheel is shown in FIG. 8. A fragrance wheel is a round diagram showing the inferred relationships among olfactory groups based upon similarities and differences in their odor. The groups bordering one another are implied to share common olfactory characteristics. Fragrance wheels are frequently used as a classification tool in oenology and perfumery.

(21) Turning now to FIG. 1, there is shown a schematic diagram that illustrates a non-limiting example of a system **100** according to an aspect of the present disclosure. In the illustrated system, electrical characteristics or conductance of the skin of a subject **102** is measured using physical collection devices **106**, such as Galvanic Skin Response (GSR) sensors, in response to a subject's exposure to a stimulus, such as a fragrance. As shown, a pair of physical collection devices **106** are placed on adjacent fingers of the subject's hand in order to measure skin conductivity between the pair of physical collection devices **106**. In some embodiments, other parameters of the subject **102** can be monitored, such as Photoplethysmogram (PPG) signals for sensing heart rate via an optional pulse probe (not shown), eye movement, body movement, facial expressions, etc.

(22) Still referring to FIG. 1, a mobile computing device **104** is coupled to the physical collection devices **106** in a wired or wireless manner to collect the skin conductance signals, such as GSR signals, generated by the physical collection devices **106**. In some embodiments, the mobile computing device **104** is used to process the collected signals, and based on the processed signals, determines a product recommendation to be presented to the subject **102**. In other embodiments, the mobile computing device **104** develops a fragrance profile based on the processed signals, from which a recommendation can be made with the assistance of, for example, a technician or fragrance consultant.

(23) In an embodiment, processing the collected GSR signals includes generating a skin conductance graph or GSR graph, such as the example shown in FIG. 9. The GSR graph can then be used by the mobile computing device **104** to recommend a product for the subject or present a fragrance profile that can aid the subject in product selection. Of course, the mobile computing device **104** can utilize the collected skin conductance signals in other ways, including via non-

graphical processing techniques, in order to provide a product recommendation to the subject.

(24) The mobile computing device **104** in other embodiments may transmit the GSR signals (processed or not) as GSR data to an optional server computing device **108** via a network **110**. In some embodiments, the network **110** may include any suitable wireless communication technology (including but not limited to Wi-Fi, WiMAX, Bluetooth, 2G, 3G, 4G, 5G, and LTE), wired communication technology (including but not limited to Ethernet, USB, and FireWire), or combinations thereof.

(25) With the GSR data received from the mobile computing device **104**, the server computing device **108** may respond back to the mobile computing device **104** with a product recommendation to be presented to the subject **102**. In other embodiments, the server computing device **108** may develop a fragrance profile from the GSR data. The fragrance profile can then be transmitted to the mobile computing device **104**. Once received by the mobile computing device **104**, a recommendation can be made with the assistance of, for example, a technician or fragrance consultant. Of course, in some embodiments, the server computing device **108** can access via the network **110** a cloud based computer processing system (not shown) to augment its processing, analyzing, generating, etc., capabilities.

(26) In some embodiments, the server computing device **108** processes the GSR data and generates a GSR graph. The GSR graph can then be used by server computing device **108** to provide a product recommendation to the mobile computing device **104** for presentation to the subject **102**. Alternatively, the generated GSR graph or the underlying data is transmitted to the mobile computing device **104** for use by the mobile computing device **104** for providing a product recommendation to the subject **102**. Again, the mobile computing device **104** or the server computing device **108** can utilize the collected skin conductance signals in other ways, including via non-graphical processing techniques, in order to provide a product recommendation to the subject.

(27) In some embodiments, the mobile computing device **104** may also be used to present an optional questionnaire to the subject **102**. The questionnaire may include questions that allow preferences of the subject **102** to be determined. In some embodiments, the questionnaire may also allow at least one personality trait to be determined. For example, in some embodiments, a personality trait can be correlated to one or more fragrance preferences, etc.

(28) In some embodiments, the questionnaire can be served to the mobile computing device **104** by the optional server computing system **108** for presentation to the subject **102**. In other embodiments, the mobile computing device **104** can generate and present the questionnaire to the subject.

(29) In some embodiments, the responses to the questionnaire are received and processed locally by the mobile computing device **104**. In other embodiments, the responses received by the mobile computing device **104** are sent to the optional server computing system **108** for processing. Of course, in some embodiments, the server computing device **108** can access via the network **110** a cloud based computer processing system (not shown) to augment its processing capabilities.

(30) In any case, processed responses to the questionnaire may be used by either the mobile computing device **104** or the server computing device **108** in conjunction with the GSR signals collected from the subject for providing, for example, a product recommendation.

(31) In some embodiments, one or more images can be captured by the mobile computing device **104** while the questionnaire is being conducted and/or during stimulus exposure. The collected images, such as video, can be analyzed for determining, for example, an emotional state of the subject **102**. In some embodiments, the results can be employed, for example, to augment the biometric data or to verify the results of the questionnaire responses when generating the product recommendation.

(32) In some embodiments, the product recommendation may be provided to the subject **102** in a convenient, simple format. For example, the product recommendation can be a specific fragrance,

e.g., identified by a tradename such as Trade Winds. Other information about the subject **102**, such as scent preferences, a personality trait, prior fragrance purchases, etc., may be also presented to the subject. Additionally or alternatively, the product recommendation may take the form of a fragrance profile. The fragrance profile can be presented as a word-based description or visually depicted as a note diagram. One example of a note diagram generated by the system **100** and presented to the subject **102** is shown in FIG. **10**. With the word-based description or the note diagram, a fragrance having a high probability of enjoyment by the subject can be selected, either independently or with the assistance of a fragrance consultant.

(33) FIG. **2** is a block diagram that illustrates various components of a non-limiting example of GSR unit **114** according to an aspect of the present disclosure. The GSR unit **114** includes a pair of physical collection devices **106** in the form of GSR sensors (“GSR sensors **106**”) and a process and transmit circuit **118**. Either dry or gel electrodes can be employed with embodiments of the present disclosure.

(34) Generally, GSR electrodes detect the changes in electrical (ionic) activity of the skin as a result from changes in sweat gland activity. The GSR sensors **106**, sometimes referred to as GSR electrodes, are configured to measure these changes, usually as skin conductance measurements, sometimes referred to as GSR signals. The GSR signals measured by the GSR sensors **106** can be suitably processed for transmission to the mobile computing device **104** for storage, data processing and/or analysis, etc. For example, the GSR signals in some embodiments are amplified by an amplifier **120** and digitized by an A/D converter **122** prior to arrival at transmitter **130**. In some embodiments, the GSR signals can be filtered in the analog domain prior to conversion by the A/D converter **122** or in the digital domain after conversion by the A/D converter **122** via one or more filters **124**.

(35) One non-limiting example of a GSR unit **114** that outputs suitable signals for use by the system **100** is the Shimmer3 GSR+ unit from Shimmer. Other GSR devices can also be used.

(36) FIG. **3** is a block diagram that illustrates various components of a non-limiting example of a mobile computing device **104** according to an aspect of the present disclosure. The mobile computing device **104** is configured to collect information from a subject **102** that reflects a physiological response (e.g., eccrine activity) based on exposure to one or more fragrances in, for example, a sequential order. In particular, the mobile computing device **104** is configured to receive GSR signals from one or more GSR sensors **106** for processing, recordation, transmission (optional) and/or analysis (optional). In some embodiments, the mobile computing device **104** is configured to receive GSR signals from the transmitter **130** of the unit **114** (see FIG. **2**).

(37) In some embodiments, the mobile computing device **104** processes the GSR signals for use in determining a product recommendation for the subject **102**. In other embodiments, as will be described in more detail below, the GSR signals (processed or not) are transmitted as GSR data to the optional server computing system **108** via the network **110** for processing and/or generating a product recommendation, etc. In either case, the mobile computing device **104** can then present the product recommendation to the subject **102**, or to a cosmetic consultant that is assisting the subject **102**. In some embodiments, generating a product recommendation can be the identification of a specific product (e.g., a specific perfume/cologne). In other embodiments, generating a product recommendation can be information, such as a fragrance profile, that assists in the selection of a specific product or product family.

(38) In some embodiments, the mobile computing device **104** may be a smartphone. In some embodiments, the mobile computing device **104** may be any other type of computing device having the illustrated components, including but not limited to a tablet computing device or a laptop computing device. In some embodiments, the mobile computing device **104** may not be mobile, but may instead be a stationary computing device such as a desktop computing device or computer kiosk. In some embodiments, the illustrated components of the mobile computing device **104** may be within a single housing. In some embodiments, the illustrated components of the mobile

computing device **104** may be in separate housings that are communicatively coupled through wired or wireless connections (such as a laptop computing device with an external camera connected via a USB cable). The mobile computing device **104** also includes other components that are not illustrated in FIG. 3, including but not limited to one or more processors, a non-transitory computer-readable medium, a power source, and one or more network communication interfaces.

(39) In order to implement some (or all) of the technology and methodology set forth herein, the mobile computing device **104** includes, in some embodiments for example, a display device **302**, a camera **304**, a GSR engine **306**, a user interface engine **308**, an optional questionnaire analysis engine **310**, an optional facial image analysis engine **312**, a recommendation engine **314**, a user data store **316** and a product data store **318**. Each of these components will be described in turn.

(40) In some embodiments, the display device **302** is any suitable type of display device, including but not limited to an LED display, an OLED display, or an LCD display, that is capable of presenting interfaces to the subject **102**. As will be described in more detail below, such interfaces include a questionnaire, a product recommendation, etc., to be presented to the subject **102**. In some embodiments, the display device **302** may include an integrated touch-sensitive portion that accepts input from the subject **102**.

(41) In some embodiments, the camera **304** is any suitable type of digital camera that is used by the mobile computing device **104** for capturing images, such as video. In some embodiments, the mobile computing device **102** may include more than one camera **304**, such as a front-facing camera and a rear-facing camera.

(42) In some embodiments, the GSR engine **306** is configured to collect GSR signals from the GSR sensors **106**, process the GSR signals, and record the GSR signals in a time-based manner as GSR data in the user data store **314**. In some embodiments, processing the GSR signals may include but is not limited to converting, filtering, transforming, and/or the like. The GSR engine **306** is also configured to process the signals in order to generate a GSR graph, such as the example shown in FIG. 9.

(43) In some embodiments, the user interface engine **308** is configured to present a user interface on the display device **302**. In some embodiments, the user interface engine **308** is configured to present a product recommendation, such as a name of a product or a fragrance profile to the user **102**. In some embodiments, the user interface engine **308** may be configured to present visualizations of the GSR data as a GSR graph on the display device **302**. In some embodiments, the user interface engine **308** may be configured to use the camera **304** to capture images of the subject **102** during the questionnaire and/or fragrance stimulus process.

(44) In some embodiments, the user interface engine **308** may be optionally configured to present on the display device **302** at least one questionnaires to the subject **102** for collecting information from the subject **102**. In some embodiments, the questionnaire aims to collect information that may be relevant to the characteristic parameters of the fragrances to which the subject was or will be exposed. For example, the questionnaire may ask via a series of true/false or multiple choice questions that may elicit preferences to certain head, middle, or base notes. For example, one question of the questionnaire, which is shown in FIG. 11, may present to the subject a number of pictures depicting scenes, such as the beach, the forest, etc., to elicit a response to which depicted scene is associated with a preferred fragrance of the subject. In another question of the questionnaire, the subject may be asked whether they prefer feminine, masculine or unisex scents. In yet another question of the questionnaire, the subject may be asked whether they prefer the scent to be perceivable, subtle, complimentary or strong. In yet another question of the questionnaire, the subject may be asked to enter her preferred fragrances, including the most recently purchased fragrance. Some or all of the collected data can be stored, for example, in the user data store **316**.

(45) In some embodiments, the questionnaire analysis engine **310** may be configured to receive responses to the questionnaire from the subject **102** via the user interface engine **308**, and may determine at least one preference, e.g., scent preference, scent characteristic, etc., of the subject **102**

based on one or more of the responses. For example, if the subject **102** chose the scene of the forest as preferred, the questionnaire analysis engine **310** may be configured to determine that the subject **102** prefers woody notes, as shown for example in the fragrance wheel of FIG. **8**. In some embodiments, the questionnaire analysis engine **310** may be configured to compare the responses to data stored, for example, in the product data store **318**. In an embodiment, the questionnaire analysis engine **310** may be configured to determine at least one personality trait of the subject **102** based on one or more of the responses.

(46) In some embodiments, the facial image analysis engine **312** is configured to analyze images (e.g., video) captured by the camera **304**. For example, in some embodiments, the facial image analysis engine **312** includes one or more image processing algorithms that analyze video captured by the camera **304** for determining facial expressions, head movement, eye tracking, etc., of the subject **102**. The images captured by the camera during use of the system **100** can be referred to as facial data.

(47) In some embodiments, the recommendation engine **314** may be configured to generate at least one product recommendation for the subject **102** based at least on the GSR data. In other embodiments, the recommendation engine **314** may be configured to generate at least one product recommendation for the subject **102** based at least on the GSR data and the optional questionnaire data. In yet other embodiments, the recommendation engine **314** may be configured to generate at least one product recommendation for the subject **102** based at least on the GSR data, the optional questionnaire data, and/or the data processed by the facial analysis engine **312**.

(48) In some embodiments, the product recommendation is in the form of a specific product, such as Trade Winds branded perfume. In other embodiments, the product recommendation is in the form of a fragrance profile. In these embodiments, the fragrance profile can be presented as a word-based description, visually depicted as a note diagram, etc. In some embodiments, the recommendation engine **314** provides the product recommendation to be presented to the subject **102** via the display device **302**.

(49) For example, FIG. **10** is one example of a fragrance note diagram that can be generated by the recommendation engine **314** and presented to the subject **102**. The note diagram depicts characteristic parameters of the fragrances preferred by the subject **102**. As shown in FIG. **10**, the note diagram depicts visually the top notes, the heart or middle notes, and optionally, the base notes, that are preferable to the subject. These notes are represented in a pattern of that form a wheel, with bar segments indicating strength of preference. For example, regarding the top notes, aromatics is depicted as five (5) bars, which the subject prefers more than floral, which is depicted as one (1) bar. Similarly, regarding the middle notes, fruity is depicted as four (4) bars, which the subject prefers more than spices, which is depicted as three (3) bars. In addition to or in the alternative to the number of bars, the color of the bars may also indicate strength of enjoyment. In some embodiments of the note diagram, the strength values may be linear (i.e., two bars is twice as preferable as one bar, etc.) or non-linear, such as Logarithmic, exponential, etc.

(50) In the diagram depicted in FIG. **10**, this subject enjoys aromatics, and to a lesser extent citrus, in the top notes, green and fruity, and to a lesser extent spices, in the middle notes, and balsamic, and to a lesser extent musk or woody, in the base notes. In some embodiments, this note diagram can be presented on the display device **302** via user interface engine **308**. In some embodiments, the note diagram can be used by a fragrance consultant to recommend a product type or product line that corresponds to the note profile of the note diagram. In other embodiments, the note diagram can be used by the subject **102** to compare to a fragrance chart of one or more fragrance sellers. In yet other embodiments, the recommendation engine **314** can analyze, for example, the image of the note diagram or the underlying data used to generate the note diagram, and based on the analysis, automatically present a product recommendation to the subject **102**.

(51) In some embodiments, the recommendation engine **314** employs one or more algorithms for analyzing the images (e.g., the GSR graph, etc.) generated from the biometric data. Based on this

analysis, one or more characteristic parameters of the fragrances preferred by the subject **102** can be determined. For example, an increase in sweat produced by the subject **102** during exposure to the fragrance(s) causes skin conductance to change. This change can be correlated to a positive stimuli (e.g. happiness or enjoyment of the fragrance), a negative stimuli (e.g. sadness or threat of the fragrance), or to the characteristic parameters (note profiles) thereof. Thus, the stimuli (e.g., fragrance) can result in an increase in arousal, and thus, in an increase in skin conductance. The GSR signals represent the intensity of the emotion felt by the subject **102** based on exposure to the stimuli.

(52) In some embodiments, the preferred characteristic parameters of the fragrance are determined based on the image(s) of the GSR data. For example, in some embodiments, image processing techniques are applied to the images in order to determine the preferred characteristic parameters of the fragrances. In some embodiments, the recommendation engine **314** may comprise or access an artificial neural network that is trained to determine the characteristic parameters based on the images. Of course, any other type of suitable machine learning technique and/or classical image processing technique may be performed in order to determine the preferred characteristic parameters of the fragrances exposed to the subject **102**.

(53) For example, in some embodiments, the recommendation engine **314** includes a machine learning model for assisting in determining the product recommendation. The machine learning model can be trained using, for example, images of GSR graph, etc., of subjects exposed to known fragrances (with known characteristic parameters, such as note profiles) that resulted in positive emotion (e.g., enjoyment) and negative emotion (e.g., dislike or sadness) by the subject **102**. In some embodiments, the images from known fragrances are used to create a set of supervised training data, and a machine learning model such as an artificial neural network may be trained with the training data using any suitable technique, including but not limited to gradient descent. The resulting machine learning model will accept an image from the GSR engine **306** as input, and will output either preferred characteristic parameters or a product recommendation that has a high probability of enjoyment by the subject **102**. In some embodiments, the preferred characteristic parameters can be used to generate a fragrance profile, such as the note diagram of FIG. **10**, of the subject **102**.

(54) In some embodiments, the processed data from the questionnaire analysis engine **310** and/or the facial analysis engine **312** can be used by the recommendation engine **314** to either augment the decisions made by the machine learning model or the signal processing carried out by the recommendation engine **314**. In some embodiments, the analyzed data from the facial analysis engine **312** can be used to verify the responses to the questionnaire analyzed by the questionnaire analysis engine **310** and vice versa.

(55) Accordingly, with the knowledge of the preferred characteristic parameters of the fragrances determined from the GSR data, and/or the questionnaire data (optional) and facial data (optional), the recommendation engine **314** is configured to determine an appropriate product stored in a product data store **318** that matches or is highly correlative to the preferred characteristic parameters determined by the system **100**. For example, the recommendation engine **314** can compare the results to a product (e.g., fragrance, etc.) map, a look-up table, etc., stored in the product data store **318**. The comparison can be based, for example, on a potential match confidence level.

(56) In some embodiments, the mobile computing device **104** may also include a user data store **316**, which is configured to store records for each subject **102** that uses the system **100**. The records may include, for example, at least one fragrance product, at least one fragrance profile, responses to a questionnaire, at least one personality trait, at least one product recommendation, facial data, and/or other information collected or determined by the system **100**. In an embodiment, feedback received from the subject **102** after having used the recommended product(s) may also be stored in the user data store **316** or forward to the product data store **318** in order to improve future product

recommendations by the system **100**.

(57) Further details about the actions performed by each of these components are provided below.

(58) “Engine” refers to logic embodied in hardware or software instructions, which can be written in a programming language, such as C, C++, COBOL, JAVA™, PHP, Perl, HTML, CSS, JavaScript, VBScript, ASPX, Microsoft .NET™, Go, and/or the like. An engine may be compiled into executable programs or written in interpreted programming languages. Software engines may be callable from other engines or from themselves. Generally, the engines described herein refer to logical modules that can be merged with other engines, or can be divided into sub-engines. The engines can be stored in any type of computer-readable medium or computer storage device and be stored on and executed by one or more general purpose computers, thus creating a special purpose computer configured to provide the engine or the functionality thereof. In some embodiments, the engines can be implemented by one or more circuits, programmable processors, application specific integrated circuits (ASICs), programmable logic devices (PLDs) and/or field programmable logic devices (FPLDs), etc.

(59) “Data store” refers to any suitable device configured to store data for access by a computing device. One example of a data store is a highly reliable, high-speed relational database management system (DBMS) executing on one or more computing devices and accessible over a high-speed network. Another example of a data store is a key-value store. However, any other suitable storage technique and/or device capable of quickly and reliably providing the stored data in response to queries may be used, and the computing device may be accessible locally instead of over a network, or may be provided as a cloud-based service. A data store may also include data stored in an organized manner on a computer-readable storage medium, such as a hard disk drive, a flash memory, RAM, ROM, or any other type of computer-readable storage medium. One of ordinary skill in the art will recognize that separate data stores described herein may be combined into a single data store, and/or a single data store described herein may be separated into multiple data stores, without departing from the scope of the present disclosure.

(60) FIG. 4 is a block diagram that illustrates various components of a non-limiting example of an optional server computing system **108** according to an aspect of the present disclosure. In these embodiments, one or more functions of the mobile computing device set forth above can be additionally or alternatively carried out by the server computing device **108**. For example, the fragrance preference information (e.g., scent preferences from, for example, biometric data (e.g., eccrine activity) from scent exposure, optional questionnaire data, and/or optional facial data) collected by the mobile computing device **104** can be transmitted, with or without additionally processing (e.g., filtering, transforming, etc.) and/or storage, to the server computing system **108** via the network **110**. In that regard, the server computing system **108** can include, for example, the GSR engine **306** (FIG. 3) for processing and storing the GSR signals and optionally generating a GSR graph, such as the example shown in FIG. 9.

(61) In some embodiments, the server computing system **108** uses the information received from the mobile computing device **104** to determine a product recommendation to be used by the subject **102**, and transmits the recommendation back to the mobile computing device **104** for presentation to the subject **102**. In that regards, the server computing system **108** can additionally or alternatively include the questionnaire analysis engine **310**, the facial image analysis engine **312**, the recommendation engine **314**, and/or the product data store **318**, the functionality of which was described in detail above. In some embodiments, the server computing system **108** may also include the user data store **316**.

(62) FIG. 5 is a block diagram that illustrates aspects of a representative computing device **400** appropriate for use as a computing device of the present disclosure. While multiple different types of computing devices were discussed above, the representative computing device **400** describes various elements that are common to many different types of computing devices, such as the mobile computing device **104** and/or the server computing device **108**. While FIG. 5 is described

with reference to a computing device that is implemented as a device on a network, the description below is applicable to servers, personal computers, mobile phones, smart phones, tablet computers, computer kiosks, embedded computing devices, and other devices that may be used to implement portions of embodiments of the present disclosure. Moreover, those of ordinary skill in the art and others will recognize that the computing device **400** may be any one of any number of currently available or yet to be developed devices.

(63) In its most basic configuration, the computing device **400** includes at least one processing circuit, such as processor **402**, and a system memory **404** connected by a communication bus **406**. Depending on the exact configuration and type of device, the system memory **404** may be volatile or nonvolatile memory, such as read only memory (“ROM”), random access memory (“RAM”), EEPROM, flash memory, or similar memory technology. Those of ordinary skill in the art and others will recognize that system memory **404** typically stores data and/or program modules that are immediately accessible to and/or currently being operated on by the processor **402**. In this regard, the processor **402** may serve as a computational center of the computing device **400** by supporting the execution of instructions.

(64) As further illustrated in FIG. 5, the computing device **400** may include a network interface **410** comprising one or more components for communicating with other devices over a network. Embodiments of the present disclosure may access basic services that utilize the network interface **410** to perform communications using common network protocols. The network interface **410** may also include a wireless network interface configured to communicate via one or more wireless communication protocols, such as WiFi, 2G, 3G, LTE, WiMAX, Bluetooth, Bluetooth low energy, and/or the like. As will be appreciated by one of ordinary skill in the art, the network interface **410** illustrated in FIG. 5 may represent one or more wireless interfaces or physical communication interfaces described and illustrated above with respect to particular components of the computing device **400**.

(65) In the exemplary embodiment depicted in FIG. 5, the computing device **400** also includes a storage medium **408**. However, services may be accessed using a computing device that does not include means for persisting data to a local storage medium. Therefore, the storage medium **408** depicted in FIG. 5 is represented with a dashed line to indicate that the storage medium **408** is optional. In any event, the storage medium **408** may be volatile or nonvolatile, removable or nonremovable, implemented using any technology capable of storing information such as, but not limited to, a hard drive, solid state drive, CD ROM, DVD, or other disk storage, magnetic cassettes, magnetic tape, magnetic disk storage, and/or the like.

(66) As used herein, the term “computer-readable medium” includes volatile and non-volatile and removable and non-removable media implemented in any method or technology capable of storing information, such as computer readable instructions, data structures, program modules, or other data. In this regard, the system memory **404** and storage medium **408** depicted in FIG. 5 are merely examples of computer-readable media.

(67) Suitable implementations of computing devices that include a processor **402**, system memory **404**, communication bus **406**, storage medium **408**, and network interface **410** are known and commercially available. For ease of illustration and because it is not important for an understanding of the claimed subject matter, FIG. 5 does not show some of the typical components of many computing devices. In this regard, the computing device **400** may include input devices, such as a camera, keyboard, keypad, mouse, microphone, touch input device, touch screen, tablet, and/or the like. Such input devices may be coupled to the computing device **400** by wired or wireless connections including RF, infrared, serial, parallel, Bluetooth, Bluetooth low energy, USB, or other suitable connections protocols using wireless or physical connections. Similarly, the computing device **400** may also include output devices such as a display, speakers, printer, etc. Since these devices are well known in the art, they are not illustrated or described further herein.

(68) In some embodiments, the plurality of sensors and/or the one or more engines form a scent

response unit. In some embodiments, the scent response unit includes processing circuitry configured to detect one of: a real-time cognitive process associated with an olfactory stimulus; a variation in an electrical characteristics of the skin associated with a response to one or more fragrance accords; detect a skin conductance based on a response to an olfactory stimulus; a change in a skin potential associated with a response to an olfactory stimulus; or conductivity fluctuations indicative of a response to an olfactory stimulus.

(69) In some embodiments, the one or more engines form a perfume selection unit that includes one of the following: processing circuitry configured to generate one or more virtual instances of a fragrance subset based on at least one input associated with an electrodermal activity measurand; processing circuitry configured to generate one or more virtual instances of a fragrance subset based on at least one input associated with a sympathetic skin response; processing circuitry configured to generate one or more instances of a degree of desirability score or a likeability measure; processing circuitry configured to generate one or more instances of a scent strength; processing circuitry configured to generate one or more instances of aromatic compound concertation; or processing circuitry configured to generate one or more instances of base notes, top notes or middle notes in a fragrance composition.

(70) FIG. 6 is a flowchart that illustrates a non-limiting example of a method for generating and/or providing product recommendations to a subject according to one or more aspects of the present disclosure. The representative method, generally designated **600**, will be described with references to one or more components of the system **100** depicted in FIGS. 1-5.

(71) Prior to starting the method **600**, one or more GSR sensors **106** are coupled to the subject **102**. In some embodiments, two GSR sensors **106** are coupled to adjacent fingers of the subject's nondominant hand. If desired, other sensors can be attached to the subject **102**, such as a heart rate sensor or the like.

(72) Once the sensors **106** have been suitably associated with a hand of a subject **102**, the method can begin. In some embodiments, as will be described below, images of the subject are captured by the mobile computing device **104** during use of the system **100**. In these embodiments, the subject **102** and the mobile computing device **104** are arranged with respect to each other such that images from camera **304** can capture at least the face of the subject **102**.

(73) From a start block, the method **600** proceeds to block **602**, where the subject **102** is exposed to one or more fragrances. For example, the subject is exposed to a sequence of fragrances. The subject **102** is exposed to each fragrance for a period of time. In an embodiment, the period of time is about 45 seconds. Of course, shorter or longer periods of time can be employed. In other embodiments, the period of time is about five (5) minutes or longer. In some embodiment, the fragrances exposed to the subject include at least two fragrance notes, such as the top and middle notes.

(74) The subject's reaction to the one or more fragrances is captured by the GSR sensors **106** and transmitted as GSR signals to the mobile computing device **104**. In that regard, the mobile device **104** is coupled in communication with the GSR sensors **106** and receives the GSR (e.g., biometric) signals of the subject **102** at block **604**. In some embodiments, the subject's reaction to the one or more fragrances is also captured by the camera **304** of the mobile computing device **104** at block **606**. Of course, the actions carried out at block **606** are optional.

(75) At block **608**, the facial image analysis engine **312** determines the emotional state of the subject **102** based on the images captured by the camera **304**. For example, the facial image analysis engine **312** employs one or more algorithms to analyze the video captured by the camera during the time periods that the subject is exposed to the fragrances (or questionnaire described below). The one or more algorithms are configured, for example, to analyze the facial expressions of the subject and/or body movement or gestures of the subject in order to determine whether or not the subject **102** enjoyed (e.g. positive emotional response) the fragrance or disliked (e.g., negative emotional response) the fragrance.

(76) In some embodiments, the video captured by the camera **304** is processed by the facial image analysis engine **312** residing on the mobile computing device **104**. In other embodiments, the video is processed by facial image analysis engine **312** residing on the server computing device **108**. In these or other embodiments, the video is stored in the user data store **316**, either locally at the mobile computing device **104** or remotely at the server computing system **108**.

(77) At block **610**, the GSR signals can be processed by the GSR engine **306** to generate, for example, GSR data. In some embodiments, the GSR signals are processed by the GSR engine **306** residing on the mobile computing device **104**. In other embodiments, the GSR signals are processed by the GSR engine **306** residing on the server computing device **108**. In these or other embodiments, the GSR data is stored in the user data store **314**, either locally at the mobile computing device **104** or remotely at the server computing system **108**.

(78) In some embodiments, the actions performed at blocks **602-610** are repeated for each fragrance to be exposed to the subject **102**. In other embodiments, the actions performed at blocks **602-606** can be performed for each fragrance prior to the actions performed at block **608** and **610**. In an embodiment, the subject is exposed to four fragrances. Of course, a number of fragrances more or less than four can be used in embodiments of the present disclosure.

(79) In one embodiment, a preselected set of fragrances are exposed to the subject **102**. For example, the preselected set of fragrances can be the top four sellers in a company's line of fragrances. In other embodiments, as will be described in more detail with regard to FIG. 7, the reaction of the subject **102** to the preceding fragrance may be used by the system **100** to influence the choice of the subsequent fragrance(s) to be exposed to the subject **102**. For example, the GSR data (e.g., the GSR graph) and/or the determined emotional state of the subject from exposure to a fragrance may be presented on the display device **302**. With the aid of a fragrance consultant, the next fragrance can be chosen for exposure to the subject **102**. Alternatively, the system **100** can be configured to automatically choose the next fragrance to be presented to the subject based on the reaction of the subject to the previous fragrance(s).

(80) In some embodiments, each fragrance exposed to the subject **102** includes at least two notes (e.g., a top and a middle note, two middle notes, a top and a base note, etc.). In some embodiments, each fragrance exposed to the subject **102** includes at least three notes (e.g., a top, a middle, and a base note (vertical accord), three middle notes (a horizontal accord), etc.) In any case, the characteristic parameters (e.g., top notes, middle notes, and/or base notes) of the fragrances exposed to the subject **102** are known and stored in the product data store **318**. In some embodiments, the characteristic parameters are stored, for example, as a note diagram.

(81) In some embodiments, a user interface engine **308** of the mobile computing device **104** optionally presents a questionnaire to the subject **102** at block **612**. In some embodiments, the questionnaire may include questions that directly represent values for the subject **102**. For example, the questionnaire may expressly ask the subject **102** to input a preference for fragrances, including specific product names, preferred notes, or other fragrance characteristics, such as whether the subject likes masculine, feminine or unisex fragrances, etc. In other embodiments, the user interface engine **308** presents one or more questions to the subject **102**, the answers of which can be used to infer the subject preferences mentioned above or others. In some embodiments, the responses from the questionnaire can be used to verify the results generated by the facial analysis engine **312** and/or the recommendation engine **314**.

(82) At block **614**, the user interface engine **308** receives responses to the questionnaire and transmits the responses to the questionnaire analysis engine **310** for processing. At block **616**, the questionnaire analysis engine **310** determines one or more fragrance preferences based on the responses to the questionnaire. In some embodiments, the responses are processed by the questionnaire analysis engine **310** residing on the mobile computing device **104**. In other embodiments, the responses are processed by the questionnaire analysis engine **310** residing on the server computing device **108**. The user interface engine **308** may receive the responses via input

into the user interface presented on the display device **302**. The responses from the questionnaire and the results from processing the responses may be stored in the user data store **316**. It will be appreciated that the actions performed at blocks **612** and **614** are also optional.

(83) At block **618**, a recommendation engine **314** determines a product recommendation based on at least the GSR data, and optionally, the preferences of the subject determined by the questionnaire engine **310** and/or the facial image analysis engine **312**. In doing so, the recommendation engine **314** may access data from the product data store **316**. In some embodiments, the product recommendation is a particular product. In other embodiments, the product recommendation is a fragrance profile. In some embodiments, the product recommendation is determined by the recommendation engine **314** residing on the mobile computing device **104**. In other embodiments, the product recommendation is determined by recommendation engine **314** residing on the server computing device **108**.

(84) At block **620**, the product recommendation is presented to the subject. For example, in one embodiment in which a specific product is presented, the product recommendation can be displayed on the display device **302** along with, for example, a description (e.g., note profile) of the product, the price of the product, where the product can be purchased, etc. In other embodiments in which the product recommendation is in the form of a fragrance profile, the fragrance profile may be displayed to a fragrance consultant by the display device **302**. With the assistance of the fragrance consultant, one or more products may be presented to the subject based on the fragrance profile.

(85) The method **600** then proceeds to an end block and terminates.

(86) FIG. **7** is a flowchart that illustrates another non-limiting example of a method for generating and/or providing product recommendations to a subject according to one or more aspects of the present disclosure. The representative method, generally designated **700**, will be described with references to one or more components of the system **100** depicted in FIGS. **1-5**. The method **700** is substantially similar to the method **600** described above with reference to FIG. **6** except for the differences that will now be described.

(87) In the embodiment of FIG. **7**, the actions of blocks **702-716** are performed for each fragrance exposed to the subject. In this embodiment, instead of the fragrances being preselected, the recommendation engine **314**, or other engine(s) of the system **100**, determines at block **718** the next fragrance to be presented to the subject **102** based on the biometric data generated from exposure to the preceding fragrance. In some embodiments, the recommendation engine **314** determines the next fragrance to be presented to the subject **102** based on the biometric data generated from exposure to the preceding fragrance and the emotional state of the subject from block **708**. In some other embodiments, the recommendation engine **314** determines the next fragrance to be presented to the subject **102** based on the biometric data generated from exposure to the preceding fragrance, the emotional state of the subject from block **708**, and one or more responses from the questionnaire from block **716**. Once all of the fragrances have been presented to the subject **102**, the method **700** then proceeds to block **720**, where the recommendation engine **314** determines a product recommendation. Of course, in some embodiments, blocks **706**, **708**, **712**, and **714**, and **716** are optional.

(88) In some embodiments, the first fragrance to be selected is determined based on one or more responses from the questionnaire. Accordingly, the questionnaire can be presented to the subject prior to any exposure to a fragrance.

(89) The detailed description set forth above in connection with the appended drawings, where like numerals reference like elements, are intended as a description of various embodiments of the present disclosure and are not intended to represent the only embodiments. Each embodiment described in this disclosure is provided merely as an example or illustration and should not be construed as preferred or advantageous over other embodiments. The illustrative examples provided herein are not intended to be exhaustive or to limit the disclosure to the precise forms

disclosed. Similarly, any steps described herein may be interchangeable with other steps, or combinations of steps, in order to achieve the same or substantially similar result. Moreover, some of the method steps can be carried serially or in parallel, or in any order unless specifically expressed or understood in the context of other method steps.

(90) In the foregoing description, specific details are set forth to provide a thorough understanding of exemplary embodiments of the present disclosure. It will be apparent to one skilled in the art, however, that the embodiments disclosed herein may be practiced without embodying all of the specific details. In some instances, well-known process steps have not been described in detail in order not to unnecessarily obscure various aspects of the present disclosure. Further, it will be appreciated that embodiments of the present disclosure may employ any combination of features described herein.

(91) The present application may also reference quantities and numbers. Unless specifically stated, such quantities and numbers are not to be considered restrictive, but exemplary of the possible quantities or numbers associated with the present application. Also in this regard, the present application may use the term “plurality” to reference a quantity or number. In this regard, the term “plurality” is meant to be any number that is more than one, for example, two, three, four, five, etc. The term “about,” “approximately,” etc., means plus or minus 5% of the stated value.

(92) Throughout this specification, terms of art may be used. These terms are to take on their ordinary meaning in the art from which they come, unless specifically defined herein or the context of their use would clearly suggest otherwise.

(93) The principles, representative embodiments, and modes of operation of the present disclosure have been described in the foregoing description. However, aspects of the present disclosure, which are intended to be protected, are not to be construed as limited to the particular embodiments disclosed. Further, the embodiments described herein are to be regarded as illustrative rather than restrictive. It will be appreciated that variations and changes may be made by others, and equivalents employed, without departing from the spirit of the present disclosure. Accordingly, it is expressly intended that all such variations, changes, and equivalents fall within the spirit and scope of the present disclosure as claimed.

Claims

1. A fragrance recommendation system, comprising: a plurality of sensors configured to sense an electrodermal response of a subject based on a response to an olfactory stimulus, wherein the olfactory stimulus is a fragrance comprising two or more notes consisting essentially of top notes, middle notes, and base notes, and wherein the electrodermal response consists of a Galvanic Skin Response (GSR) signal; a user interface; and processing circuitry connected in communication with the plurality of sensors and to the user interface, the processing circuitry configured to: receive a first GSR signal of the subject based on a first presented fragrance consisting of at least two fragrance notes; process the first GSR signal to generate a first GSR data; generate, based on the first GSR data, a recommended second presented fragrance consisting of at least two fragrance notes; receive a second GSR signal of the subject based on the recommended second presented fragrance; process the second GSR signal to generate a second GSR data; generate, based on the first GSR data and the second GSR data, one or more instances of predicted base notes, top notes, or middle notes in the first and second presented fragrances; generate, based on the first GSR data and the second GSR data, one or more instances of a predicted degree of desirability score in the first and second presented fragrances; generate a product recommendation based on the generated instances of predicted base notes, top notes, or middle notes, and the predicted degree of desirability score; and provide the product recommendation to the subject via the user interface, the product recommendation including one or more of a specific brand name of a fragrance or a visual representation of a fragrance profile.

2. The system of claim 1, wherein the processing circuitry is housed in a mobile computing device.
3. The system of claim 1, wherein the generated GSR data is represented as an image.
4. The system of claim 1, wherein the processing circuitry is configured to determine a product recommendation by comparing the data indicative of the generated instances to product data accessible by the one or more engines.
5. The system of claim 1, wherein the fragrance profile is presented to the subject as the product recommendation.
6. The system of claim 1, wherein the product recommendation is generated by comparing the fragrance profile with a set of fragrance profiles representing, respectively, a set of fragrances, the set of fragrance profiles accessible by the one or more engines.
7. The system of claim 1, further comprising a camera configured to capture data indicative of the emotional state of the subject based on the response to the olfactory stimulus, wherein the product recommendation is generated based at least on said GSR data and said data indicative of the emotional state of the subject.
8. The system of claim 1, wherein the processing circuitry is configured to detect the electrodermal response based on the response to an olfactory stimulus includes processing circuitry configured to: detect a real-time cognitive process associated with an olfactory stimulus; detect a variation in an electrical characteristics of the skin associated with a response to one or more fragrance accords; detect a skin conductance based on a response to an olfactory stimulus; detect a change in a skin potential associated with a response to an olfactory stimulus; or detect conductivity fluctuations indicative of a response to an olfactory stimulus.
9. A method for recommending a fragrance to a subject, comprising: exposing the subject to a first presented fragrance consisting essentially of base notes, middle notes, and top notes, the subject having one or more Galvanic Skin Response (GSR) sensors coupled thereto; obtaining a first GSR data of a subject from the one or more GSR sensors based on exposure to the first presented fragrance; analyzing the first GSR data; based on the first GSR data, generating by processor circuitry a recommended second presented fragrance consisting essentially of base notes, middle notes, and top notes; exposing the subject to the second presented fragrance; obtaining a second GSR data of the subject from the one or more GSR sensors based on exposure to the second presented fragrance; analyzing the second GSR data; based on at least the first GSR data and the second GSR data, generating by the processor circuitry one or more instances of predicted base notes, top notes, or middle notes in the fragrances and one or more instances of a predicted degree of desirability score; generating, by the processor circuitry or other processor circuitry, a fragrance recommendation based on at least the generated instances of predicted base notes, top notes, or middle notes, and the predicted degree of desirability score; and presenting the fragrance recommendation to the subject.
10. The method of claim 9, wherein presenting the fragrance recommendation to the subject includes presenting a fragrance name to the subject based on a fragrance profile that comprises two or more notes that are preferred by the user; or presenting the fragrance profile to the subject.
11. The method of claim 9, further comprising obtaining questionnaire data of the subject indicative of a preference of a characteristic parameter of a product, wherein said fragrance recommendation is based on the analyzed biometric data and the questionnaire data.
12. The method of claim 10, wherein the fragrance recommendation is a name of a perfume.
13. The method of claim 10, wherein said generating a fragrance recommendation includes generating a fragrance profile of notes preferred by the user based on the GSR data, comparing the fragrance profile with a set of fragrance profiles representing, respectively, a set of fragrances to select a fragrance from the set of fragrances having a fragrance profile most similar to the generated fragrance profile.
14. The method of claim 9, further comprising obtaining data indicative of the emotional state of the subject based on the exposure to the fragrance, wherein said fragrance recommendation is

based on the analyzed GSR data and said data indicative of the emotional state of the subject.

15. The method of claim 9, wherein presenting the fragrance recommendation to the subject includes displaying, by a display of a computing device, the fragrance recommendation to the subject.
